7

SDG 2 and the Dominance of Food Security in the Global Agri-food Norm Cluster

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The world is facing severe levels of food insecurity and environmental degradation related to agri-food practices. World hunger is increasingly facilitated by the negative impacts of the COVID-19-pandemic and the consequences of the Russian invasion of Ukraine. Between 702 and 828 million people are affected by hunger (FAO et al, 2022: 10). Since 2015, the UN's Agenda 2030 for Sustainable Development has aimed to end world hunger and achieve food security, improved nutrition and sustainable agriculture through the SDG 2 (Zero hunger) (UN, 2015a). However, according to the High Level Panel of Experts on Food Security and Nutrition to the Committee on World Food Security (CFS), it is impossible to accomplish this goal by 2030 without radical governance transformation (HLPE, 2020).

Global agri-food governance typifies current sustainability challenges and is well suited to inform our knowledge on global norm stability and change. Even though SDG 2 as a key component of Agenda 2030 serves as the commonly accepted global normative reference for agri-food governance, different agri-food norms compete with each other, and ongoing norm debates hamper the fight against hunger (Breitmeier et al, 2021a; 2021b). Moreover, agri-food norms are linked to a range of environmental issues, and there are several synergies and trade-offs between SDG 2 and the 'green goals' (SDGs 6, 13–15) of Agenda 2030 (Griggs et al, 2017). While, on the one hand, the agri-food sector is a major driver of environmental pollution, on the other hand, agri-food production systems are prone to negative environmental changes with effects on global food security (Rockström et al, 2009).

Building on research on global norms and norm clusters in international relations (eg Lantis and Wunderlich, 2018), and on agri-food governance and norm development (Clapp and Scott, 2018; Breitmeier et al, 2021a; 2021b; Clapp et al, 2022), this chapter reflects on the normative foundations of SDG 2. After laying out the theoretical framework, the chapter traces different historical phases of norm development in global agri-food governance since the Second World War (see Table 7.1).

Food security is at the core of a norm cluster in addition to improved nutrition and sustainable agriculture. As a core norm, food security perpetuates approaches that are primarily designed to increase agri-food production and technological innovations and do not inherently acknowledge environmental considerations of sustainability. A subsequent section assesses how SDG 2 reflects the historically grown cluster of global agri-food norms. The goal formally integrates different agri-food norms, while the dominance of food security continues to hinder environmentally salient governance approaches. Alternative policy actors, such as organic and food sovereignty movements, have not yet succeeded in promoting sustainable agriculture.

7.1 Norm clusters and global agri-food governance

Norms define what can be considered appropriate behaviour. The content of a global norm can be shaped and localized in political discourses and thus varies between different institutional settings, cultural practices and over time (Acharya, 2004). A *norm cluster* is understood as a collection of 'aligned, but distinct, norms or principles that relate to a common, overarching issue area; they address different aspects and contain specific normative obligations' (Lantis and Wunderlich, 2018: 571). Such a norm cluster can consist of a mix of old norms that have received social weight over time, for example by their application in practice or their formalization in international law (Fehl and Rosert, 2020), and new norms that emanate from new and politicized norm debates (Sandholtz, 2023). Change in norm clusters can occur either when the substantial content of an already established norm is modified, for example by including new normative elements, or when new norms are institutionalized and constitutionalized next to established ones (Lantis and Wunderlich, 2018).

Whether a norm is being strengthened, maintained, or weakened depends on the degree to which policy actors advance or support pro-norm arguments in norm contestation (Sandholtz, 2023). Public and private *norm entrepreneurs* can raise awareness of new norms, establish new ways of talking about and understanding issues, reframe a formerly unproblematic phenomenon to become problematic, and attempt to convince a critical mass to embrace newly established norms (Finnemore and Sikkink, 1998: 895). Norm entrepreneurship in agri-food governance relates to the agency of consumers and producers (individuals or groups), that is, their capacity to make self-determined decisions, for example, about what foods they eat or produce, how that food is distributed within food systems and their ability to engage in processes that shape agri-food system policies (HLPE, 2020: 9). Norm entrepreneurs politicize or depoliticize agri-food issues by constituting or removing an issue area from the public agenda, respectively (Breitmeier et al, 2021a). Such politicization renders ongoing norm contestation visible (Wiener, 2014).

The politicization of a norm cluster triggers normative clarification and refinement processes on both procedural and substantive grounds (Lantis and Wunderlich, 2018). It also provokes the development of alternative norms and goals that not only facilitate the replacement of old norms with new ones but can also foster contextual sense making (Zimmermann, 2016) and the discursive reframing and development of established norms (Breitmeier et al, 2021a).

Norm contestation in the agri-food sector is accompanied by asymmetric power structures that enable donor countries (such as the US and the UK), international institutions (such as the FAO), and transnational business actors (such as Bayer or Nestlé) to shape the agenda in accordance with their interests and norm understandings (Breitmeier et al, 2021b). The advancing globalization of food markets over the past three decades has put business actors in a position to influence the broad lines of agri-food research and development and to make governance decisions themselves, for example, by setting private labelling standards (Clapp and Scott, 2018). Even though policy actors from civil society have succeeded in raising alternative normative frames in norm contestation, such as La Via Campesina promoting *food sovereignty*, they usually have fewer resources compared to corporate actors and therefore tend to have less agency (Clapp and Scott, 2018; McKeon, 2021).

Breitmeier et al (2021a: 626) have applied the norm cluster concept to the agri-food sector and highlight that policy actors display different interpretations of sustainability, prioritizing certain elements of a norm cluster and neglecting others. In this vein, food movements politicize old norms, promote alternative normative frames and thereby redirect policy foci (Lang and Barling, 2012: 317; McKeon, 2021: 49). Their emergence leads to changes in norm prioritization and fuels new norm dynamics. In this vein, as the global agri-food norm cluster is characterized by ongoing contestation, there is also an ongoing debate about appropriate governance approaches (Breitmeier et al, 2021a; Schwindenhammer, 2023). However, as outlined in the following sections, despite ongoing norm contestation and new links between old and new norms and principles, the food security norm continues to be the most robust one in the global agri-food norm cluster. Environmental aspects have been left behind.

7.2 Historical norm development in global agri-food governance

This section traces different and cumulating historical phases of norm development in global agri-food governance (see also Table 7.1). While, in addition to food security, norms for nutrition security and sustainable agriculture have emerged, food security continues to be the primary norm and it perpetuates production- and technology-oriented agri-food approaches. As these approaches neglect detrimental environmental and social impacts, they have given rise to alternative policy actors, such as organic and food sovereignty movements, that promote normative counter-frames in global agri-food governance (see Figure 7.1).

7.2.1 The emergence of the food security norm

The global agri-food norm cluster is essentially characterized by the fight against hunger and dominated by the food security norm. After the Second World War, the right to food was embodied in the 1948 UN Universal Declaration of Human Rights. It was then codified in international human rights law in the International Covenant on Economic, Social and Cultural Rights in 1966 (Khoo, 2010).

Figure 7.1: Normative counter-frames

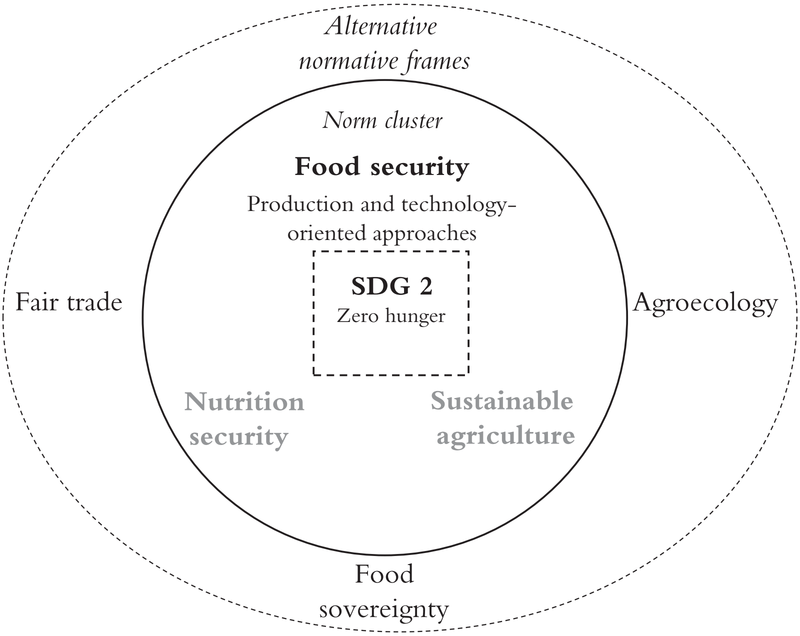

Table 7.1: Historical milestones in global agri-food governance

1945	Foundation of FAO
1948	Universal Declaration of Human Rights embodies the right to food
Since 1950s	Green Revolution (industrialization of agriculture)
1966	International Covenant on Economic, Social and Cultural Rights (ICESCR) codifies the RTF in international human rights law (Article 11)
Since 1970s	Fair trade movements (eg Worldshops)
1972	Foundation of International Federation of Organic Agriculture Movements (IFOAM)
1974	World Food Conference: foundation of Committee on World Food Security (CFS)
1993	Foundation of La Via Campesina (food sovereignty movement)
1996	World Food Summit establishes 'right to adequate food'; Rome Declaration establishes four pillars concept of food security and envisages halving the proportion of people who suffered from hunger up to 2015
1997	Foundation of Fairtrade International (originally Fairtrade Labelling Organizations International)
2000	Millennium Declaration establishes MDG 1 to 'eradicate extreme poverty and hunger' by 2015
2009	Reform of the CFS (institutional participation of non-governmental actors) and establishment of High Level Panel of Experts on Food Security and Nutrition as science-policy interface of the CFS
2012	UN Summit on Sustainable Development (Rio+20) connects food and nutrition security to sustainable agriculture
2014	Second International Conference on Nutrition
2015–2030	Agenda 2030 and SDG 2 (Zero hunger)

Article 11 of the ICESCR recognizes 'the fundamental right of everyone to be free from hunger' and outlines individual and collective policy measures, including specific programs:

> (a) To improve methods of production, conservation and distribution of food by making full use of technical and scientific knowledge, by disseminating knowledge of the principles of nutrition and by developing or reforming agrarian systems in such a way as to achieve the most efficient development and utilization of natural resources; (b) Taking into account the problems of both food-importing and food-exporting countries, to ensure an equitable distribution of world food supplies in relation to need. (UN, 1966: 4)

The ICESCR directed international agri-food governance towards ending hunger by means of increasing global food production, trade and aid. But the wording of Article 11 of the ICESCR captured early on the tension between the fundamental human right of everyone to be free from hunger and the policy focus on controlling food surpluses and deficits, partly driven by the big powers' self-interested aid and trade objectives (Khoo, 2010: 37).

McKeon (2015) identifies three milestones of post-Second World War agri-food governance: the creation of the FAO in 1945, the World Food Conference in 1974 and the reform of the CFS in 2009. The FAO is the UN agency that has led international efforts in global agri-food governance for several decades now. The first World Food Conference was held in Rome in 1974 under the auspices of the FAO, in the wake of the devastating famine in Bangladesh over the preceding two years. The conference sought to establish methods to help poor countries finance food purchases, to induce rich countries to provide capital and technical aid to help the developing countries improve domestic production, and to create an international grain reserve system to prevent local famines (Biwas and Biwas, 1975: 20). In the same year, the CFS was established as an intergovernmental body to serve as an additional forum in the UN system to review and follow up on policies concerning world food security including production and physical and economic access to food (Duncan, 2015).

By that time, in the 1970s, food security was understood as a national concept, with the nation-state being food secure when there was sufficient food available for its citizens (Roberts, 2021: 64). The 1996 World Food Summit fundamentally changed this concept and shifted the focus from nation-states to individuals (Roberts, 2021: 65). With the 'right to adequate food', the summit delegates launched a rights-based approach to food security (Roberts, 2021: 65). They defined food security as existing 'when all people, at all times, have *physical and economic access* to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life' (FAO, 1996; emphasis added).

With the Rome Declaration on World Food Security and a related Plan of Action, delegates formulated the goal of reducing world hunger by half no later than the year 2015. According to the Rome Declaration, food security rests on four pillars: food availability, access, utilization and stability (FAO, 1996). With this concept, it was increasingly acknowledged that food insecurity is a problem of structural poverty, markets and market structure, and the relative affordability of different types of food rather than only an issue of food availability (Battersby and Crush, 2016). However, the Plan of Action was unable to reconcile the market-oriented approach to food security prioritizing global agri-food trade, which 'deepened an agrarian crisis in the Global South among small-scale farmers, who had lost price supports and food subsidies via Structural Adjustment loan conditions' (Canfield et al, 2021: 4).

With the advancing globalization of agri-food trade since the 1990s, North/South asymmetries intensified in the agri-food sector, with implications on human development. The majority of undernourished people live in the Global South. The volatile global prices of agricultural commodities, especially cash crops such as coffee and cocoa, often force farmers to sell their products at below production costs. This, compounded by smallholders' lack of access to social infrastructure and services and frequently insecurity of land tenure, leads to further disadvantages (Matthews, 2015). Asymmetries also persist owing to subsidies for domestic agriculture, especially in the EU and the US, and to trade distortions through colonial heritage at the nation-state and international levels (Battersby and Crush, 2016). Global South groups have already raised issues of North/South asymmetries in international agri-food markets since the 1970s. So-called Worldshops (or Third World or Fair Trade shops) started to sell 'fair trade' commodities. Diverse national labelling organizations joined forces and established the Fairtrade Labelling Organizations International, now called Fairtrade International, to harmonize standards in 1997 (Barratt Brown, 2007).

Once again, reaffirming the significance of food security for global agri-food governance, the 2000 UN Millennium Declaration referred to food security in Millennium Development Goal 1, which was aimed at eradicating extreme poverty and hunger. Like the 1996 Rome Declaration, target 1.3 envisaged halving the proportion of people who suffered from hunger until 2015, compared to the baseline year of 1990. In 2009, in response to global food price spikes in 2007–8, the Declaration of the World Summit on Food Security officially reaffirmed an extension of the 1996 Rome definition of food security by adding the word 'social' to the phrase 'physical, social and economic access' (CFS, 2012: 5). In the same year, the food security governance space institutionally broadened with the reform of the CFS. The committee was opened up to the participation of non-state actors, such as NGOs, research institutions and representatives of philanthropic foundations. The CFS was meant to turn into '*the* international platform for the discussion and coordination of food security policy', striving for a world free from hunger (Duncan, 2015: 86). Moreover, the 2012 UN Conference on Sustainable Development (Rio+20) reaffirmed 'the right to an adequate standard of living, including *the right to food*' (UN, 2012: 2; emphasis added). However, in 2015, MDG 1.3 was narrowly missed as the proportion of undernourished people in countries of the Global South fell by just under half, from 23.3 per cent in 1990–92 to 12.9 per cent in 2014–16 (UN, 2015b).

7.2.2 The spread of production- and technology-related approaches

The food security norm has been and still is the primary norm underlying global agri-food governance (Breitmeier et al, 2021a). It is closely connected to the historical emergence of a production-oriented approach that aims at enhanced agricultural productivity through intensified large-scale agricultural production relying on monoculture, high-yield varieties of grains, intensive use of synthetic fertilizers (Lu and Tian, 2017: 181) and on-field application of biotechnology, for example the cultivation of crops with resistance to herbicides, pest attacks and glyphosate (Perry, 2016).

A variety of norm entrepreneurs, such as agri-food corporations, international organizations, foundations, bilateral donors and university researchers, have 'helped forge a global food system that is increasingly specialised, dependent on trade, and premised on the need to produce more food with industrial methods – all in the name of improving efficiency' (Clapp and Moseley, 2020: 1408). The so-called Green Revolution, which implemented the production-oriented approach, was a turning point for agri-food systems across many countries in the 1950s (Leach et al, 2020: 7). Agricultural subsidies in industrialized countries further fuelled the intensification of agricultural production (Breitmeier et al, 2021a: 631). Modern agri-food production following the production-oriented approach has been credited for increasing the amount of agricultural output over time (Holt-Giménez, 2011: 316), and global food crises strengthened the approach, most recently, the 2007–8 food crisis (Fouilleux et al, 2017), the COVID-19-pandemic (McKeon, 2021: 6) and the war against Ukraine (Ben Hassen and El Bilali, 2022).

The production-oriented approach is closely linked to approaches promoting technology innovation in the agri-food sector. While, back in the 1950s, the industrialization of agriculture was supposed to increase food production volumes, new technologies today are meant to pave the way for visions of circular and high-tech agri-food systems (Schwindenhammer, 2019; 2023). In recent years, central policy documents such as FAO reports (eg FAO et al, 2022) and the SDG framework (to be discussed later) have stressed the potential of new technologies to transform agri-food systems. In consequence, the spread of research, development and application of synthetic biotechnology, field robotics, drone, sensor and nutrient recovery technologies can be observed (Nuijten et al, 2019; Schwindenhammer, 2020).

However, the production- and technology-oriented approaches were and continue to be basically implemented regardless of environmental and social concerns. The production-oriented approach has actually caused a number of negative impacts, such as air pollution, soil acidification and degradation, water eutrophication, biodiversity loss, the monopolization of seed and chemical inputs by companies from the Global North resulting in raising costs for farmers, and the displacement of millions of peasants to fragile hillsides, shrinking forests and urban slums (Holt-Giménez, 2011: 316; Lu and Tian, 2017: 181–2). In response to these impacts and market distortions, alternative normative frames and trajectories evolved, especially in the late 1960s and 1970s. Social movements, such as La Via Campesina (Spanish for 'The Peasants' Way') since the 1990s, have promoted normative counter-frames (Holt-Giménez, 2011). In particular, the norm of food sovereignty emphasizes people's agency by claiming the right to define their own food systems to ensure their own livelihoods and access to culturally appropriate foods (Clapp et al, 2022: 4). Rather than the result of insufficient volumes, hunger is considered to be caused by privileging access to food, and the movement demands the redistribution of control over systems of production and consumption (Holt-Giménez et al, 2021).

7.2.3 The norm of nutrition in addition to food security

Irrespective of movements' counter-frames, the norm of *nutrition security* emerged in the mid-1990s to add to the dominant norm of food security, and established a new phase of global agri-food governance (El Bilali et al, 2019). The new norm extends and contextualizes the human rights approach to food security by specifying human nutritional and social needs (El Bilali et al, 2019: 3). Among diverse organizations, the Bill and Melinda Gates Foundation stands out as a norm entrepreneur regarding this new norm development phase. The world's largest philanthropic foundation promoted the norm by initiating the Global Alliance for Improved Nutrition in 2002, which has successfully endorsed fortified foods and multivitamins to address malnutrition and undernutrition for two decades now (Moravaridi, 2012).

In 2012 the CFS recommended defining food and nutrition security as something that:

> exists when all people at all times have physical, social and economic access to food, which is safe and consumed in sufficient quantity and quality to meet their dietary needs and food preferences, and is supported by an environment of adequate sanitation, health services and care, allowing for a healthy and active life. (CFS, 2012: 8)

The 2012 *State of Food Insecurity in the World* report stresses that economic and agricultural growth should be 'nutrition-sensitive' (FAO et al, 2012: 20). In 2014, after the Second International Conference on Nutrition in Rome, the HLPE identified five key issues of agri-food policy transformation: (1) healthy nutrition in changing food systems; (2) challenges and opportunities of livestock systems, food security and nutrition; (3) inequalities and food security and nutrition referring to the imperative of addressing the needs of disadvantaged and vulnerable populations; (4) the increasing role of financial markets in food security and nutrition; and (5) pathways to sustainable food systems in the pursuit of human and environmental health for all (HLPE, 2014: 6).

7.2.4 The 'new' norm of sustainable agriculture

In addition to the norms of food and nutrition security, the norm of *sustainable agriculture* has emerged in global agri-food governance and established a third central phase of global agri-food governance since the Second World War (Lang and Barling, 2012). The agri-food norm cluster now also covers issues of environmentally sound practices with regard to the use of land, water, fertilizers and other resources; economic efficiency in input costs; crop productivity; and farm income (Mockshell and Kamanda, 2017). It also refers to socially adequate conditions and impacts of agricultural practices with regard to knowledge preservation, livelihood improvement, distribution of land ownership, and intra- as well as intergenerational equity (Mockshell and Kamanda, 2017).

Sustainable agriculture has a long history. The International Federation of Organic Agriculture Movements was an essential norm entrepreneur. It was founded in 1972 with the aim of stopping the expansion of industrialized agriculture (Schwindenhammer, 2017). Industrialized agriculture and its detrimental impacts on the environment originally provoked a normative counter-frame of 'agroecology'. This frame includes replacing synthetic fertilizers, pesticides and hybrid seeds, which are common to industrialized agriculture, with natural manures, cover crops and animal-based fertilization. Moreover, organic movements aim for more direct producer–consumer relations. They want to reduce farmers' financial risks such as vulnerability to price volatility (Holt-Giménez et al, 2021). Some alternative local networks even try to overcome consumer/producer dichotomies, which are inherent in the global capitalist order. For example, in community-supported agriculture (CSA), people contribute membership fees instead of paying a price per food item (Peuker, 2015).

Since the late 1980s, the norm of sustainable agriculture has gained momentum in the context of the global discourse on sustainable development. The Brundtland Report provided a first conceptual definition of sustainability, which stresses the interconnection between the three dimensions of environmental, economic and social sustainability, and postulates intra- and intergenerational justice (WCED, 1987). In the 2000s growing global concern about the negative environmental impacts of agri-food production based on the production-oriented approach led to intensified attempts of agri-food policy institutions to capture sustainability (Canfield et al, 2021: 6). The 2012 UN Summit on Sustainable Development (Rio+20) officially connected food and nutrition security to sustainable agriculture, and emphasized the need to promote more sustainable agriculture 'that improves food security, eradicates hunger and is economically viable, while conserving land, water, plant and animal genetic resources, biodiversity and ecosystems and enhancing resilience to climate change and natural disasters' (UN, 2012: 22).

FAO included sustainable agriculture in the Strategic Framework 2000–2015 and added agro-ecological intensification to the reviewed Strategic Framework in 2017 (FAO, 2017). After FAO (2014) had already pursued the water–energy–food (WEF) nexus approach, which conceptualizes water, energy and food as inextricably linked and dynamically interacting, since 2014, the reviewed 2017 framework also points to crop diversification, ecosystem service and decent living conditions for rural people and small-scale farmers (FAO, 2014; 2017; Breitmeier et al, 2021a). Finally, in 2018 HLPE (2020: 8) suggested extending the four-pillar concept of food security to a six-pillar concept by incorporating the two dimensions of *sustainability* and *agency* to 'codify what is already incorporated in international legal guidance on the right to food' (Clapp et al, 2022: 8). By doing so, HLPE took up the demands of the food sovereignty movement for more self-determined agri-food systems in addition to environmental concerns.

7.3 SDG 2: squaring the circle of the agri-food norm cluster

Agenda 2030 connects the global norms of food security, improved nutrition and sustainable agriculture in SDG 2 (UN, 2015a). This section outlines how the goal reflects the agri-food norm cluster in global governance. The international community has formally addressed different agri-food norms, but food security still dominates and impacts the realization of new norms such as sustainable agriculture. This robustness of the food security norm perpetuates production- and technology-oriented approaches that set aside environmental issues such as water pollution and climate change.

SDG 2 consists of five targets and three subtargets (see Table 7.2). The first five targets (SDG 2.1–2.5) are directly related to food security and agricultural sustainability, while the last three subtargets (SDG 2a–2c) are market-related measures aimed at increasing agricultural investments and reducing market distortions and price volatility (Gil et al, 2019: 686). Behind each target are indicators by which to track progress. Indicators are stipulated as a way of monitoring and uniformly testing progression of the goals across differing urban landscapes (Merino-Saum et al, 2020).

The ranking of targets corresponds to the historical emergence of the norms and the rise of production- and technology-oriented approaches, which were described earlier. Target 2.1 aims to 'end hunger and ensure access by all people … to safe, nutritious and sufficient food all year round (by 2030)'. Hence Agenda 2030 first and foremost reinforces norms of food security and improved nutrition by strengthening the goals defined by the

Table 7.2: Sustainable Development Goal 2 (Zero hunger)

2.1 By 2030, end hunger and ensure access by all people, in particular the poor and people in vulnerable situations, including infants, to safe, nutritious and sufficient food all year round.
2.2 By 2030, end all forms of malnutrition, including achieving, by 2025, the internationally agreed targets on stunting and wasting in children under 5 years of age, and address the nutritional needs of adolescent girls, pregnant and lactating women and older persons.
2.3 By 2030, double the agricultural productivity and incomes of small-scale food producers, in particular women, indigenous peoples, family farmers, pastoralists and fishers, including through secure and equal access to land, other productive resources and inputs, knowledge, financial services, markets and opportunities for value addition and non-farm employment.
2.4 By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, that help maintain ecosystems, that strengthen capacity for adaptation to climate change, extreme weather, drought, flooding and other disasters and that progressively improve land and soil quality.
2.5 By 2020, maintain the genetic diversity of seeds, cultivated plants and farmed and domesticated animals and their related wild species, including through soundly managed and diversified seed and plant banks at the national, regional and international levels, and promote access to and fair and equitable sharing of benefits arising from the utilization of genetic resources and associated traditional knowledge, as internationally agreed.
2.A Increase investment, including through enhanced international cooperation, in rural infrastructure, agricultural research and extension services, technology development and plant and livestock gene banks in order to enhance agricultural productive capacity in developing countries, in particular least developed countries.
2.B Correct and prevent trade restrictions and distortions in world agricultural markets, including through the parallel elimination of all forms of agricultural export subsidies and all export measures with equivalent effect, in accordance with the mandate of the Doha Development Round.
2.C Adopt measures to ensure the proper functioning of food commodity markets and their derivatives and facilitate timely access to market information, including on food reserves, in order to help limit extreme food price volatility.

Source: UN (2015a).

1996 Rome Declaration and the 2000 Millennium Declaration. Target 2.2 emphasizes the need to end *malnutrition* in particular. However, unlike the 2012 UN Summit on Sustainable Development (Rio+20), there is no official mention of the human right to food (Vivero Pol and Schuftan, 2016; see also Mehta et al, Chapter 6 in this volume).

Instead of linking 'zero hunger' to problems of structural poverty, markets and market structure, and the relative affordability of different types of food (Battersby and Crush, 2016), SDG 2 falls back on a narrow problem definition of food availability in conjunction with production- and technology-oriented approaches. Target 2.3 aims to 'double the agricultural productivity'. Incomes of small-scale food producers are mentioned but not linked to distributional issues and structural asymmetries (including colonial heritage). Following the production-oriented approach, it is suggested that 'productive resources and inputs, knowledge, financial services, markets and opportunities for value addition and non-farm employment' could compensate for persistent disadvantages.

In line with the norm of sustainability agriculture, target 2.4 mentions the need to ensure systems and practices that 'increase (agricultural) productivity and production, that help maintain ecosystems, that strengthen capacity for adaptation to climate change, extreme weather, drought, flooding and other disasters and that progressively improve land and soil quality'. Because of this target, Gupta and Vegelin (2016: 441–2) consider SDG 2 to be environmentally relevant (see also Partzsch, Chapter 1 in this volume). Demonstrating links between organic and sustainability agriculture, Germany indicates the share of organic farming land in the country as the (only) indicator to track progress on target 2.4 (9.7 per cent in 2021) (DeStatis, 2023). However, other countries such as France (Cling et al, 2019), the UK (UK Government, 2022) and the US (US Government, 2022), are still 'exploring data sources' and do not report on this target at all. This lack of a commonly agreed indicator demonstrates the deferral of the environmental subtarget in global sustainability governance.

Finally, the fifth target as well as three additional subtargets (2.A, 2.B and 2.C) reveal a deep belief in the powers of the market to manage to achieve zero hunger. By contrast, again, there is no mention of historically grown asymmetries and disadvantages of small-scale producers in the Global South. Instead, in line with the production- and technology-oriented approaches, there is the aim to 'increase investment … in … technology development and plant and livestock gene banks in order to enhance agricultural productive capacity in developing countries, in particular least developed countries'.

In sum, SDG 2 opens up new perspectives for integrated and cross-sectoral policies (Sachs et al, 2019), but at the same time demonstrates contested norms in global agri-food governance. With the fourth subtarget, theoretically, Agenda 2030 requires transforming traditional policies in such a way that ecosystem protection is acknowledged to be a prerequisite for food security (Breitmeier et al, 2021a). However, unlike especially the first target on food security, there is as yet no commonly used indicator to even monitor the fourth target (Gil et al, 2019: 685). In consequence, SDG 2 is likely to invoke multiple synergies and trade-offs with the green goals of Agenda 2030. In particular, as mentioned earlier, SDG 2 has high negative interactions with SDG 6. Moreover, agri-food production competes with the expansion of energy infrastructure (SDG 7) for land and water (Fader et al, 2018; Dabla and Goldthau, Chapter 8 in this volume).

7.4 Conclusion

Building on research on global norms in international relations and agri-food governance, this chapter reflects on the normative foundations of SDG 2. It points to current sustainability challenges, and contributes to research on global norm stability and change, conflicting perceptions and alternative frames in global agri-food governance. As this chapter has shown, the global agri-food norm cluster consists of aligned, but distinct, historically evolved norms and approaches that collide and are continuously contested. Food security turns out to be the core of the global agri-food norm cluster, in addition to improved nutrition and sustainable agriculture. The normative priority of global agri-food governance is to ensure food security following production and technology-oriented approaches. This means that the international community aims to increase production volumes on the basis of technological innovation but without regard for the environmental consequences and the social origins of hunger. The robustness of the food security norm, which has even been strengthened by the COVID-19-pandemic and reinforced by the Russian invasion of Ukraine, hence has detrimental environmental and social impacts, which continue to exist with SDG 2.

As has been shown, SDG 2 refers to the global agri-food norm cluster and formally connects the norms of food security, improved nutrition and sustainable agriculture. However, the food security norm implies that production- and technology-oriented approaches continue to dominate, and therefore the norm collides with other agri-food norms, especially sustainable agriculture. SDG 2 also invokes multiple trade-offs with the green goals of Agenda 2030, in particular SDG 6 and SDG 13. Hence, to implement SDG 2, there is an urgent need to transform the current global agri-food systems. Future agri-food policies and solutions have to consider the environmental and social impacts of agriculture and food issues against the backdrop of existing norm collisions and norm contestations in the global agri-food norm cluster.

Conventional agri-food systems have been shown to be unable to respond to broader contextual demands including environmental change. In consequence, alternative normative frames such as food sovereignty and agroecology have increasingly gained in popularity. They have entered norm contestation, but this norm contestation is characterized by the unequal representation of different norms. Alternative norm entrepreneurs, such as organic and food sovereignty movements, advocate alternatives that add to but do not replace the dominant norms and practices.

All in all, this chapter's findings substantiate the need for further research on the impact and resolution of the collision of old and new norms in agri-food governance and on how alternative actors, knowledge and developments can enter global norm contestation to promote greater environmental protection as well as social justice. An ideationally enriched research perspective allows for the capture of the conditions that warrant equal representation of different types of knowledge in norm contestation. For the phase after 2030, global agri-food governance must be prevented from leaving behind those who already have a disadvantaged voice in agri-food systems, especially local actors and social movements that have long advocated for environmentally salient agri-food systems.

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Interview with Sandra Schwindenhammer: Partnering with the Enemy to Achieve SDG 2?

Laura Kräh and Ruth Krötz

Agro-ecology is seen as an alternative approach to achieving SDG 2 but is seldom practised. Why is the transition so hard and which dependencies need to be contested for its realization?

Schwindenhammer: The SDGs reflect a general normative reference framework. While deciding on sustainability goals in general is easy for state representatives, the implementation and measurement of concrete targets is much more challenging and contested. SDG 2 aims at combating world hunger by increasing food availability and, while this is important, conventional approaches still widely neglect the social and environmental dimensions of the global hunger challenge. We face various complex problems that cannot be solved by more production and increased technology alone. Rather, the voices of farmers from the Global South need to be heard more. Agroecology and localized approaches to agri-food production can help to ensure self-sufficiency, but smallholder farmers do want not only to produce for themselves but to be equally treated as production partners. The global agri-food market creates unequal outcomes as there are subsidies and large corporations that influence the broad lines of how the food system is organized. For instance, smallholder farmers from the Global South who want to enter the European organic market face several access barriers. It must become easier for them to enter the market and we need standards that fit their situation. Farmers from the Global South face specific social and environmental challenges; barriers to agro-ecology in Germany and Rwanda are not the same.

What kind of governance and cooperation is needed to contest the trade-offs with green SDGs?

Schwindenhammer: Governance actors, especially state representatives, need to be aware of these trade-offs. There is a need for more integrated governance and discourse between different government departments to overcome the sectoral logic of policy making. Civil society actors face the impacts of the trade-offs on the ground, but also have an important role in making agri-food policy challenges visible: we need civil society's voice to bring attention to the issue. We also need to end the blame game: while state representatives call for the business sector to transform, the business sector demands that market policies be changed, but who is responsible? The truth is that all groups need to contribute.

So is there a possibility of getting companies on board for sustainable agriculture?

Schwindenhammer: Business actors have emerged as political actors and influence today's agri-food policies and governance frameworks. From a critical perspective, the rise of corporate norm entrepreneurship limits the influence and authority of states to set and implement rules. But there is hope! Getting companies on board allows us to hold them more accountable and to take advantage of their resources to develop and implement private standards. However, what counts is that solutions are sustainable, no matter whether they are implemented by Nestlé, Kraft or environmental organizations so partnering with the enemy is an entry point. Nevertheless, the question of greenwashing needs to be continually addressed.